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Chapter 1

The Planning Process



Introduction

This section deals with Master Production Scheduling (MPS) and Material Requirements Planning (MRP). The two applications share a great deal of common logic, which is described in this section. The logic which is specific to each application is described in subsequent sections.

MPS/MRP scheduling/planning runs are a series of calculations using data derived from within Style Production and from external applications such as Style Sales Order Processing, Style Forecasting and Style Purchase Management, to construct a statement of future production requirement.



This can only be done when planning models and associated reporting profiles have been defined; and the current sales, purchases and forecast data entered

The output from the runs is a series of suggestions which will:

- Suggest new production and purchase orders to meet required demand.
- Re-plan any existing orders, by suggesting changes to order quantities and due dates.

Although both MPS and MRP utilise the same processing rules, the distinction is that MPS is designed to give a high level review of demand for sales and critical (to production) items; and the MRP run extends that plan to include the demand for all materials. In addition, MPS provides additional processing facilities to enable capacity planning.

Common Processes



The MPS and MRP runs generate data specific to a planning model; the planning model defines the stockrooms to consider

The MPS/MRP runs comprise two distinct stages:

- Pre-processing.
- Processing.

Once the pre- and processing stages have been completed you must review the output produced and make scheduling and procurement decisions based on the recommendations.

Pre-processing

This stage accumulates the reference data for the processing stage. The stage may be further broken down into a number of steps:

- Step 1 - Low level code analysis.
- Step 2 - Item selection & stock consolidation.
- Step 3 - Extract sales orders.
- Step 4 - Extract production orders.
- Step 5 - [MRP] Extract demand from MPS.
- Step 6 - Extract purchase orders.
- Step 7 - Extract sales and stock forecasts.

Processing

This stage provides the analysis and produces the recommendations; and it too can be broken down into a number of phases:

- Phase 1 - Consumption of forecast.
- Phase 2 - Calculate daily demand.
- Phase 3 - Adjust supply to meet net demand.
- Phase 4 - Summarise variant details to style level.
- Phase 5 - Explode recommendations.
- Phase 6 - Generate daily machine/labour loading requirement.

Run Parameters

MPS and MRP have their own, independent run submission screen (Run Parameters screen). These screens determine the scope of the run, utilising the following defined dates and periods shown in the example and defined below.

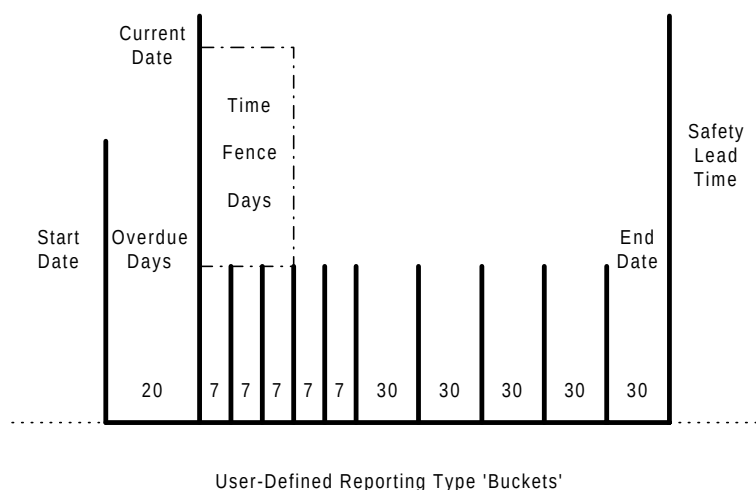


Figure -: MPS and MRP Run Dates

Current Date

This is the live date for the plan. The start date and frozen period (Time Fence Days) are calculated from this date.

Overdue Days

Indicates the number of days of overdue supply and demand you wish to consider.

Start Date

Not displayed on MRP run parameters. This is the first date which is considered for the run. Orders prior to this date are ignored. This is calculated as:

Current date *less* Overdue days

Time Fence Days

The period between the current date and the time fence date is regarded as fixed; that is, a frozen period (schedule) in which no recommendations can be made. This may be one of the following:

- Entered directly.
- Production lead time for each style.

The second option is actioned by setting the Use Item Time Fence parameter to 1 (yes) The production lead time is held on the item definition, Production Details screen; if no time has been entered or calculated, the run will use the global, default Time Fence Days setting.

Safety Horizon Days

This ensures that the end date is extended so that all generated demand is included in the lower level analysis.

End Date

This is the last date to be considered in the run. This may be one of the following:

- Entered directly *plus* Safety lead time.
- Cumulative lead time for each style *plus* Safety lead time.

The second option is actioned by setting the Use Item End Date parameter to 1 (yes). The cumulative lead time is held on the item definition, Production Details screen; if no time has been entered or calculated, the run will use the global, default end date.



In the calculation of Time Fence Days and End Date for non-production items, the purchasing lead time is used in place of production and cumulative lead times

Pre-Processing

Both MPS and MRP execute an explosion of high level demand into lower level supplies and demands.

- Demand. An anticipated event that reduces the stockroom inventory of an item, eg
 - Issues to satisfy customer sales.
 - Issues to satisfy production orders.
- Supply. An anticipated event that increases the stockroom inventory of an item, eg
 - Receipt from a production order.
 - Receipt from a purchase order.



Sales forecasts also represent demands, because if the forecast is accurate and can be satisfied, then the inventory of the forecasted product will decrease.

Style Production Planning applications all the demands and supplies on all of your items and identify where you need to change your plans. This is achieved by calculating the likely stock level of every item at every point in time within a given period which is under your control. At each point, the stock level is compared with the ideal stock of the item (which needs to be specified when defining an item). Where the likely stock is outside your parameters, the system will inform you of a suggested action.

In order to operate effectively, these routines must first of all consider the demand on your finished products. The system can then determine whether your current plans will provide enough stock of the products at the correct time. If this is not the case, then it will recommend suggested supplies, which will usually be production orders.

All existing production orders and new suggestions will create corresponding demands on component items. So when the finished products have been reviewed, the components can then be reviewed in a similar manner.

This process will normally continue until all items have been reviewed, although the system provides many different ways of operating to suit your own particular needs.

MPS and MRP both comprise two stages:

- Pre-processing. Identifies and extracts all the supplies and demands to be reviewed in the run. These are written to a temporary work area in which they can be processed.
- Processing. The sequenced review, processing and recommendations previously described.

Within these stages and beyond, references are made to three types of demand for an item:

- Independent demand.
- Dependent demand.
- Actual demand.

Independent demand

Independent demand is demand that has been input directly against the item itself. It is most commonly associated with sales orders or forecasts.

Independent demand does not involve any item other than that to which it relates. It is demand that cannot be calculated or derived but that must originate from an external source.

Dependent demand

Dependent demand is demand on one item that results from a plan to produce another. If you plan to produce (or 'supply') a product then you must issue materials to make it. Such anticipated issues constitute dependent demand. This is because the demand on the material depends on the demand to supply the product.

Broadly speaking, the dependent demand (that is, issue of the materials) precedes its corresponding supply (that is, the receipt of the production order).

There may be any number of dependent demands associated with a single supply, or production order, because there is one for every material. Note that if the production order for a product is cancelled, all of the corresponding dependent demands disappear because there are no longer any issues anticipated against the order.

For example, if you manufacture shoes, including all components, then you will normally have a sales and/or forecast demand for your shoes.

If you manufacture the soles of your shoes as a separate process, they will have their own route/BOM.

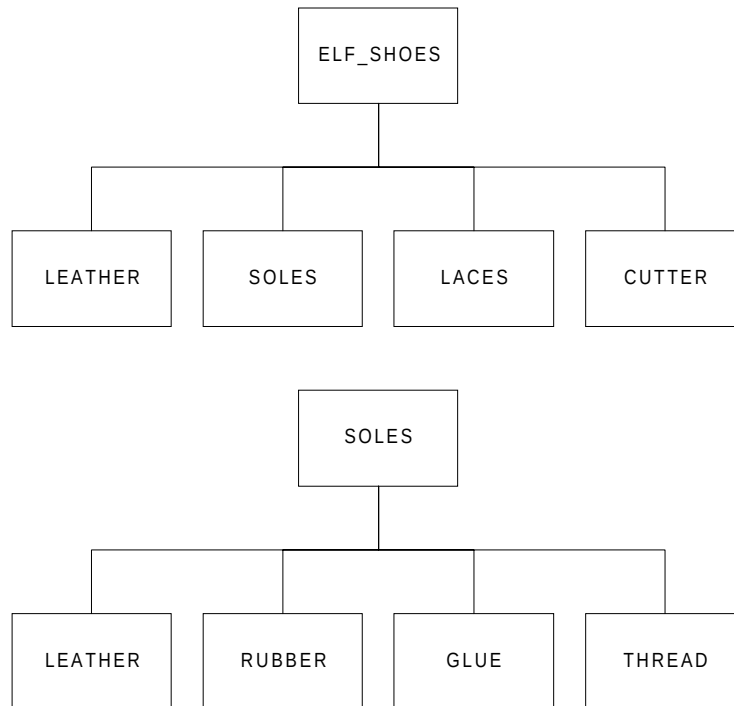


Figure -: Shoe Manufacture

ELF_SHOES will have independent demand which will need to be balanced by a supply (production order). This will create a dependent demand for SOLES which will need to be balanced by a supply (production order). It will also generate a dependent demand for LACES which will need to be balanced by a supply (purchase order).

This will create a dependent demand for RUBBER which will need to be balanced by a supply (purchase order).

However:

- LACES may be sold independently and so be subject to independent demand in the form of sales orders or forecasts as well as the dependent demand.
- Purchase orders for RUBBER may be placed well in advance, and therefore be controlled by independent demand in the form of stock forecasts as well as dependent demand.

Actual Demand

Actual demand is simply the sum of independent and dependent demand on an item.

Step 1: Low Level Code Analysis

This step carries out checks on the database to ensure that no conditions are present which will cause the planning run to fail. Such checks are carried out here, rather than when the conditions are created on the database, for performance reasons.

The step comprises the following two routines:

- Low level code integrity checking
- Assignment of low level codes



The *Low Level Code Generation* routine on the Style Production, Utilities menu may be run prior to running MPS and MRP in order to validate route/BOMs. This will also enable you to safely skip this phase which will have the added benefit of shortening the run time.



If the Suppress Low Level Code Generation field on the Run Parameters screen is set to 1 (yes) this phase will be skipped. The default depends on whether or not the system has detected changes to planning routes.

[MPS] The Lowest Level to Review field on the Run Parameters screen determines how many levels within route/BOMs to search for occurrences of items.

If you have multi-level route/BOMs then it is essential that you realise the possible consequences of setting a level which might not include the lowest level at which an item occurs.

Low Level Code Integrity Checking

The run examines all planning route route/BOMs and amended standard production order route/BOMs for recursions in structures. That is, a parent/child relationship which is logically inconsistent with other defined relationships. For example:

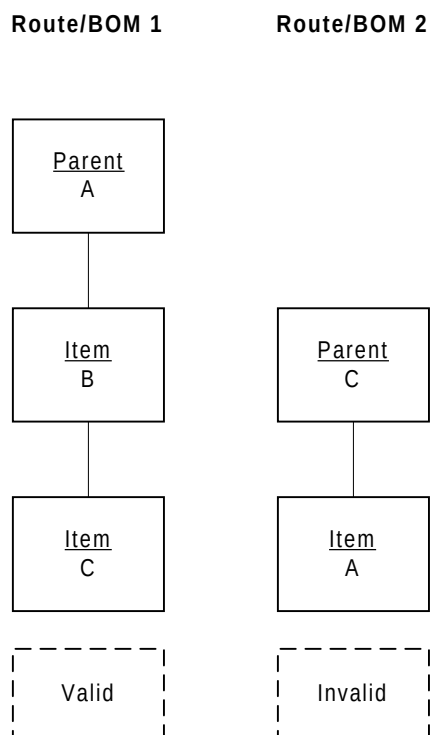


Figure -: Route/BOM Validation

Route/BOM 2 is invalid as 'A' is a parent of 'C' in route/BOM 1. Logically, an item cannot be both the child and the parent.



Such occurrences will generate an exceptions report showing all invalid parent/child relationships and the MPS or MRP run will terminate

You will then need to correct your route/BOMs before re-submitting the job

Assignment of Low Level Codes

The low level code of an item is a field which is held on the Item Master to indicate the lowest level at which the item appears in any route/BOM. The main sequence in which the planning routines review items is based on the low level code. This mechanism ensures that all higher level dependent demand on an item has been generated before that item is reviewed.

During the maintenance of the route/BOM the lowest position of an item might change. Therefore, to ensure correct processing of the planning routines they normally commence with a review of the low level codes. Any changes are updated on the Item Master.



The lowest level code may be reviewed by looking at the Item Master enquiry on the Production Definition Management, Enquiries & Reports menu

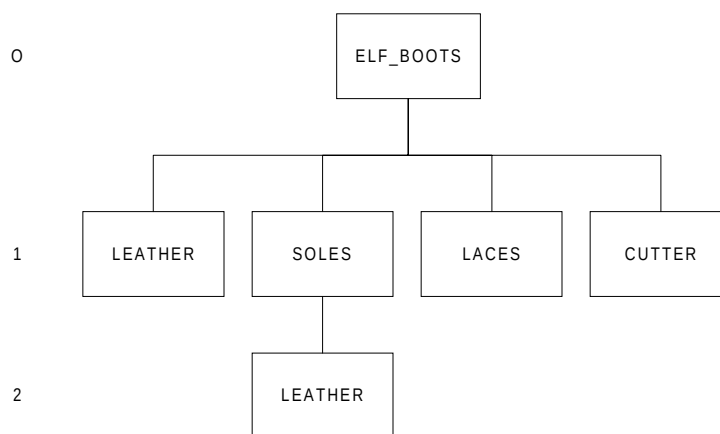


Figure -: Assignment of Low Level Codes

Items with no parents in route/BOMs are given a low level code of zero (0). Items with one level of parent are set at level 1, and so on. In the example, LEATHER appears at levels 1 and 2, so that the low level code will be 2.



It is unusual for route/BOMs within the Style Production environment to create levels lower than 2

Step 2: Item Selection And Stock Consolidation

This step reviews all items on the Item Master and extracts those which satisfy the parameters specified on the Run Parameters screen, and those which will be required to appear on review screens.



An item is defined as MPS or MRP by the setting of the Planning Type flag on the item definition, which may be set to 0 (MRP) or 1 (MPS). Broadly speaking, MPS items are selected and reviewed within MPS, whilst MRP items are selected and reviewed within MRP. There are however exceptions, which are discussed where relevant.

This means that manufactured items are normally classed as MPS items, whereas purchased items are usually classed as MRP items.

Item Selection

The review selects items which meet the following criteria:

- [MPS] MPS items with a low level code less than or equal to the specified low level code cut-off which have a stockroom balance in one or more of the stockrooms defined in the planning model
- [MRP] MRP items residing in stockrooms defined in the planning model
- Items which are not in the planning model but have dependent items that meet the above criteria
- [MRP] MRP items which meet user selection criteria if an MRP Selective run has been invoked
- [MRP] MRP items dependent on selectively reviewed items
- [MPS] MRP items with a low level code less than the selected cut-off level which have MPS items as components, with low level codes less than or equal to the cut-off level. This is done to ensure that all possible dependent demand for MPS items is trapped. However, this type of structure is possibly undesirable and assessment of the MPS status of such items should be made

Stock Consolidation

Stock balances from Style Inventory Management in each valid stockroom are accumulated as physical stock (less frozen stock) and a consolidated stock record is created for each item, from the style/colour/size records. This figure forms the opening balance on MPS/MRP reviews.

To obtain the lead time, Style uses the Item/Supplier profile for the first Supplier code in the Primary Stockroom balance file. If this does not exist, then the lead time on the Stockroom record is used.

The header enquiry in MPS review shows the first supplier code on the Stockroom balance file.

When Requisitioning extracts the suggested Purchase orders, it looks to the Item/Supplier profile for suppliers defined on the stockroom balance file and picks up the first one that exists. If none of these exist, then it will pick up the first one it can find on the file for the item. If there isn't one then it will use the first supplier code on the stockroom balance file - but will highlight this on the Reqs review. If there are not any supplier codes on the stockroom balance file then it will leave the supplier code blank on the reqs file and highlight this in the reqs review.

Step 3: Extract Sales Orders

This routine selects the outstanding sales order balances for the items selected in Step 2. Every order line which satisfies the following criteria is summarised for the item/colour/size record extracted:

- The order status is blank.
- There is an outstanding quantity to be shipped.
- The shipment date is within the start and end date of the run.
- The (warehouse) stockroom/depot is defined in the planning model.

Step 4: Extract Production Orders

[MPS] Within MPS this routine selects:

- Supplies associated with the MPS items selected in Step 2.
- Dependent demands on items selected in Step 2 emanating from supplies of MRP items.

[MRP] Within MRP this routine selects:

- Supplies associated with the MRP items selected in Step 2.
- Dependent demands on items selected in Step 2 emanating from supplies of MPS items.

Each order is checked to see if it matches the following criteria:

- The due date is within the start and end date of the run.
- The receiving stockroom of the order is contained in the planning model. If the stockroom is not in the model, the order is exploded down to its material (component) demand. (For [MPS], this is dependent on the low level code specified for the run.)

The materials are checked for being a selected item (Step 2) and having an issuing stockroom in the planning model; and a required date within the start and end dates of the run. This ensures that all relevant generated demand is identified

The outstanding supply quantity of the selected production orders is calculated as the order quantity less the quantity received.

The requirement quantity, dependent demand, of materials for the orders is calculated as the quantity required less the quantity issued to the orders.

Step 5: Extract Demand from MPS (MRP)

This routine is only present in MRP. It is not necessary in MPS. Its purpose is to extract dependent demand which was generated in MPS but is applicable to relevant MRP items.

It reviews every suggested order for an MPS item and determines if it has a dependent demand for an MRP item selected in Step 2. If so, the demand record is checked to establish the issuing stockroom. If it is used in the (MRP) planning model the demand is extracted for review. If the material requirement date is outside the run start date and end date, it is not considered. The outstanding requirement quantity is calculated as the quantity required less the quantity issued.

If the Include Suggested MPS (production orders) parameter for the run is set to 1 (yes) the corresponding MPS planning model is reviewed and suggested MPS production orders extracted. The MRP run uses the designated planning route of the MPS parent item to identify the MRP item material requirement. The MPS suggested due date and quantity are used as the basis of the requirement calculation, providing the time-phased MRP item demands.

Effectivity checks are made to establish whether the MRP materials are valid for use at the time of the requirement (see Phase 4 of processing)

Step 6: Extract Purchase Orders

Purchase orders are extracted in two stages:

- Extract scheduled and blanket orders
- Extract outstanding orders

Extract Scheduled & Blanket Orders

Scheduled and blanket orders are extracted if the deliver-to stockroom is defined in the planning model. The uncommitted quantity outstanding on the schedule is planned for receipt at the end of the run. This will enable the recognition of uncommitted portion of the schedule for possible rescheduling to meet earlier shortages.

Extract Outstanding Orders

This extracts both outstanding orders and goods received but not yet booked into stockrooms. Outstanding (both standard and cut-offs against schedule) orders are extracted where the due (receipt) date is within the start and end date of the run. the receipt date represents the latest of:

- Due date.
- Expected date.
- Promised date.

Only purchase orders for common materials, material group items, finished goods and subcontracted items will be extracted.

Goods inwards and inspection status purchase orders are extracted. These orders are not available for rescheduling.

The outstanding quantity on extracted orders is taken from Style Purchase or Inventory Management.

Step 7: Extract Sales And Stock Forecasts

Sales and stock forecasts for items within the planning model are extracted.



Forecast data will only be extracted if item forecasts have been generated via the Forecast Management routines or transferred from the Style Forecasting application

The degree of processing involved at this stage depends on the levels of forecast generated for each item:

- If variant level only forecasts exist, then a style level forecast will be generated automatically.
- If variant level forecasts do not exist, but the style level forecasts do, then these forecasts will be flagged as 'Style Level Only'. In this case, processing routines will not include them in the netting phase. Additionally, the Phase 3 processing stage will generate a 'Style Level Only' supply order (production or purchase).
- If style/colour and style/colour/size but not style level forecasts exist, a style level forecast will be generated from the style/colour/size forecast.

Forecasts which are at 'Style Level Only' will be identified by a demand status code prefixed by a 'Y' For example, YF STYLE forecast.

The forecast extraction/generation process may be summarised in the following table:

Forecast Extraction/Generation Summary				
Style	Style/ Colour	Style/ Colour/ Size	Extract	Generate Style From
X	X	✓	Style/colour/ size	Style/colour/size
X	✓	X	Nothing	Style/colour
✓	X	X	Style	
X	✓	✓	Style/colour/ size	Style/colour/size
✓	X	✓	Style Style/colour/ size	
✓	✓	X	Style	
✓	✓	✓	Style Style/colour/ size	
✓ - Forecasts		X - Forecasts do not exist		

exists

Processing

The MPS/MRP run reviews selected items, as described in the pre-processing stage; and in the case of [MPS] finishing at the low level code selected on the run parameters.

The review operates in the following logical phases, which run recurrently until all items are processed.

- Consumption of forecast.
- Calculate daily net demand.
- Adjust supply to meet net demand.
- Summarise variant details to style level.
- Explode recommendations.
- Generate daily machine/labour loadings.

Phase 1: Consumption of Forecast

The MPS/MRP Consume Forecast flag on the Company Profile, Planning Options screen determines whether this comparison is made as a:

- Discrete comparison.
- Cumulative consumption.

The comparison is further governed by the:

- Demand policy parameter setting on the item definition, Production Details screen, which may modify the consumption process for a particular or range of items.



If forecasting is not required for an item (as determined by its demand policy), then the demand will originate solely from sales orders and the cumulative consumption of forecast will be forced. If forecasting is required, forecasts must have been generated via the Forecast Management routines.

Discrete Comparison

This method compares the forecast for the forecast period with the actual demand in the same period. This is therefore a 'discrete' period by period direct comparison. Unless subsequently modified by the item demand policy, the greater of these will be used as the demand figure for the item for the run. The discrete method is only used where a sales forecast exists as an element of demand; if a forecast is not present, the cumulative method will be forced. For example:

Discrete Comparison of Forecast					
	Period Start				
	1/10/95	8/10/95	15/10/95	22/10/95	TOTAL
Forecast demand	100	100	100	100	400
Actual demand	150	90	30	110	380
Period Demand	150	100	100	110	460

Cumulative Consumption

This method compares the cumulative forecast demand to the end of each forecast period and the cumulative actual demand to the same date.

Unless modified by the item demand policy, demand to be used in the run will be calculated in each period as the greater cumulative figure at the end of the period minus the calculated (consumed) demand at the beginning of the period.

Such demand is calculated as a series of 5 operations, described in the following table.

Cumulative Consumption of Forecast	
	Period Start

	1/10/95	8/10/95	15/10/95	22/10/95	TOTAL
Forecast	100	100	100	100	400
Actual demand	150	90	30	110	380
(Op1) Forecast cumulative	100	200	300	400	
(Op2) Actual demand cumulative	150	240	270	380	
(Op3) Demand cumulative. Set the demand cumulative as the greater of forecast cumulative or actual demand cumulative	150	240	300	400	
(Op4) Consumed cumulative. Sum the consumed demand cumulative, excluding the current period	0	150	240	300	
(Op5) Period demand. Calculate the demand for the current period:	150	90	60	100	400

Demand Policy

The consumption of forecast calculation is dependent on the demand policy set on the item definition, Production Details screen, which defines the actual demand calculation for each period. Demand may comprise one or more of the three potential elements:

- Sales demand
- Forecast demand
- Dependent demand

Demand Policies		
Code	Description	Demand Definition
0 or blank	Total actual demand	Sum of sales orders <i>plus</i> dependent demand; no forecast
1	Forecasts compared with independent demand	Greater of (forecast <i>or</i> sales orders) <i>plus</i> dependent demand
2	Forecasts compared with total demand	Greater of (sales orders <i>plus</i> dependent demand) or forecasts
3	Independent demand compared with dependent demand	Greater of sales orders <i>or</i> dependent demand; no forecast
5	Make to forecast	Equal to forecast only; no sales orders or dependent demand
6	Total demand	Total demand; ie sum of forecasts <i>plus</i> sales orders <i>plus</i> dependent demand

Demand Policies & Consumption Of Forecast

The interaction between demand policies and consumption of forecasts is shown on the following examples:

Discrete Comparison <i>without</i> Dependent Demand										
Periods		1	2	3	4	5	6	7	8	Total
A) Dependent Demand										
B) Cumulative										
C) Forecast		10	50	90	10	30	80	20	10	300
D) Sales orders		20	90	40	50	10	20	80	50	360
E) Forecast cumulative										
F) Sales cumulative										
Demand Policy	Calculation									
0	D + A	20	90	40	50	10	20	80	50	360
1	> [D or C]	20	90	90	50	30	80	80	50	490
2	> [D + A] or C	20	90	90	50	30	80	80	50	490
3	> [D or A]	20	90	40	50	10	20	80	50	360
5	C	10	50	90	10	30	80	20	10	300
6	D + A + C	30	140	130	60	40	100	100	60	660

Figure -: Discrete Comparison of Forecast Without Dependent Demand

Discrete Comparison <i>with</i> Dependent Demand										
Periods		1	2	3	4	5	6	7	8	Total
A) Dependent Demand		10	100	20	50	60	10	100	30	
B) Cumulative										
C) Forecast		10	50	90	10	30	80	20	10	300

D) Sales orders		20	90	40	50	10	20	80	50	360
E) Forecast cumulative										
F) Sales cumulative										
Demand Policy	Calculation									
0	D + A	20	190	60	100	70	30	180	80	740
1	> [D or C] + A	30	190	110	100	90	90	180	80	870
2	> [D + A] or C	30	190	90	100	70	80	180	80	820
3	> [D or A]	20	100	40	50	60	20	100	50	440
5	C	10	50	90	10	30	80	20	10	300
6	D + A + C	40	240	150	110	100	110	200	90	1040

Figure -: Discrete Comparison of Forecast with Dependent Demand

Cumulative Consumption <i>without</i> Dependent Demand										
Periods		1	2	3	4	5	6	7	8	Total
A) Dependent Demand										
B) Cumulative										
C) Forecast		10	20	90	10	0	200	20	20	370
D) Sales orders		90	20	0	90	80	30	20	50	380
E) Forecast cumulative		10	30	120	130	130	330	350	370	
F) Sales cumulative		90	110	110	200	280	310	330	380	
Demand Policy	Calculation									
0	(F + B) - p	90	20	0	90	80	30	20	50	380
1	>[F or E] - p	90	20	10	80	80	50	20	30	380

2	$>[F + B] \text{ or } E - p$	90	20	10	80	80	50	20	30	380
3	$>[F \text{ or } B] - p$	90	20	0	90	80	30	20	50	380
5	$E - p$	10	20	90	10	0	200	20	20	370
6	$F + B + E - p$	100	40	90	100	80	230	40	70	750
Where 'p' is the previous period's consumption										

Figure -: Cumulative Consumption of Forecast without Dependent Demand

Cumulative Consumption <i>with</i> Dependent Demand										
Periods		1	2	3	4	5	6	7	8	Total
A)	Dependent Demand	10	100	20	50	50	10	100	3	370
B)	Cumulative	10	110	130	180	230	240	340	370	
C)	Forecast	10	50	90	10	30	80	20	10	300
D)	Sales orders	20	90	40	50	10	20	80	50	360
E)	Forecast cumulative	10	60	150	160	190	270	290	300	
F)	Sales cumulative	20	110	150	200	210	230	310	360	
Demand Policy	Calculation									
0	$(F + B) - p$	30	190	60	100	70	30	180	80	740
1	$>([F \text{ or } E] - p) + A$	30	190	60	100	60	70	140	80	730
2	$>[F + B] \text{ or } E - p$	30	190	60	100	70	30	180	80	740
3	$>[F \text{ or } B] - p$	20	90	40	50	40	10	100	30	380
5	$E - p$	10	50	90	10	30	80	20	10	300
6	$F + B + E - p$	40	240	150	110	100	110	200	90	1040
Where 'p' is the previous period's consumption										

Figure -: Cumulative Consumption of Forecast with Dependent Demand

Phase 2: Calculate Daily Net Demand

The run calculates for each selected item:

- Gross demand.
- Available Stock.
- Net Demand.

On a daily basis.

Gross Demand

Netted sales forecasts, sales orders and dependent production order demand

plus

Safety stock (target inventory level)

plus

Stock forecast

Available Stock

Physical stock (for the first period in the run)

Or

Projected available stock (for future periods in run)

plus

Outstanding production or purchase order quantities

Net Demand

Gross demand *less* Available stock

Three conditions may arise:

- Net demand < 0
- Net demand > 0
- Net demand = 0

Phase 3: Adjust Supply to Meet Net Demand

Having calculated the net demand for the day, the run will ensure that any imbalance is rectified by making recommended changes to supply orders. There are a number of factors which have an influence on recommendations:

- Time fence.
- Planning filter.
- Order policy codes.

Net Demand < 0

If net demand is negative, this indicates an over supply condition. The run will recommend a reduction in supply quantity, or a cancellation of an order if the demand is non existent.

Net Demand > 0

If net demand is positive, this indicates a supply shortage. The run will attempt to cover the shortage by rescheduling 'in' or 'out' any 'Cancelled' orders first then reschedule 'live' orders; it will also recommend a quantity change if necessary.

As a last resort, it will generate a suggested supply order(s) to meet the deficit.

New order quantities are dependent on the order policy code set on the item definition, Production Details screen.

Net Demand = 0

If net demand is zero, this indicates that supply and demand is in balance and the current plan does not need modification. No further action is required.

Daily Net Demand Recommendations Review Summary	
Net Demand < 0	Reduce or cancel order if due on day
Net Demand > 0	1. Increase order quantity if due on day 2. Reschedule order (possibly change quantity) 3. Suggest new order
Net Demand = 0	No recommended changes

Time Fence Days

No recommendations will be made to change orders in timing or quantity within the time fence days; to re-schedule orders into the frozen period; or to suggest new orders due inside this period. The time fence period is considered frozen by the run and the schedule inside it will remain fixed.

Planning Filters

Rescheduling of MPS/MRP recommendations can be controlled by the use of planning filters. Planning filters are defined on the parameter file (type WTYP) and can be accessed from the Style Production, Utilities menu. Default values

may be set for an item via the item definition, (Production Details screen) or for a specific production order.

Planning filters control the MPS/MRP rescheduling of production order dates and/or quantities.

Date reschedule policy - Production orders may be expedited (bring due date 'in', schedule earlier) or delayed (deferred, take due date 'out'). Date rescheduling recommendations can be restricted as follows:

- No reschedule allowed.
- Any reschedule allowed.
- Reschedule allowed if not less than a specified number of days.
- Reschedule allowed if not less than a specified percentage of the lead time.
- Reschedule allowed as long as the resulting value of the order is greater than the extended value.

where:

- Order value = Suggested qty x Inventory standard cost

Quantity reschedule policy - Production order quantities may be increased, decreased or cancelled. Quantity rescheduling recommendations can be restricted as follows:

- No reschedule allowed.
- Any reschedule allowed.
- Reschedule allowed if not less than minimum order quantity.
- Reschedule allowed if not less than the entered percentage of an item's safety stock.

The operation of the system may be modified globally by changing the rules for standard order types, or it may be modified to vary according to specific situations using the planning filter.

Such modifications could, for example, be used to inhibit all change to production orders which are 'Active', that is, for which work-in-process has been reported; and to treat all 'Planned' and 'Released' status orders as confirmed and not subject to recommendations.

Recommendations for the cancellation of an order may also be either allowed or restricted. However, cancellation is necessary when rescheduling 'out' (ie deferring).

These modifications are available not only for production orders using the planning filter, but also for purchase orders. Different rules may govern the scheduling of normal, schedule, blanket and user-defined purchase orders.



Although it is possible to define recommendations for changes to suggested orders; SP (suggested purchase) and SW (suggested production order); such recommendations will be ignored

The planning filter assigned to an item or individual order conditions the run response. If the reschedule policy is defined both for the order type and for the assigned filter, then the rules defined on the filter will be used. Where a filter is not used, the rules applying to the order type will be used. If neither order type nor filter has a policy defined, the order is not subject to recommendation.

For additional information on planning filters, see the Style Production Definition Management product guide, Production Company Definition, Parameter Files.

Order Policy Codes

The item definition, Production Details screen, includes an order policy code and associated data fields which provide parameters for the suggested orders.

Order Policy Codes				
Code	Description	Recommended Quantity	Example	
A	Discrete	Required quantity.	Actual demand Recommended qty	100 100
B	Discrete above minimum	Required quantity unless this is less than the specified <i>Minimum Order Quantity</i> in which case the minimum order quantity will be suggested.	Minimum order qty Actual demand Recommended qty	200 100 200
D	Fixed quantity	Enough batches of the <i>Fixed Order Quantity</i> to satisfy the required quantity.	Fixed qty Actual demand Recommended qty	80 100 160
G	Number of days supply	Sufficient quantity to satisfy all requirements for the specified <i>No. of days Supply</i> from the initial shortage date.	No. days supply Monday Tuesday Wednesday Recommended qty	3 20 20 30 70
H	Multiples above minimum	Enough batches of the <i>Multiple Order Quantity</i> to satisfy demand; unless this is less than the <i>Minimum Order Quantity</i> when batches equal to the minimum order quantity will be suggested.	Minimum order qty Maximum order qty Multiple order qty Actual demand Recommended qty Actual demand Recommended qty Actual demand Recommended qty	200 500 50 101 200 380 400 501 550

	Multiples up to a maximum	Maximum would equal physical maximum possible in a factory.	Maximum order qty	500
			Multiple order qty	50
			Actual demand	101
			Recommended qty	200
			Actual demand	380
			Recommended qty	400
			Actual demand	501
			Recommended qty	500
			Actual demand	50
			Recommended qty	

Phase 4: Summarise Variant Details to Style Level

Processing in the first 3 Phases is performed at style/colour/size level, whereas Phase 4 updates at style level. The summarising is different dependent on the type of record being processed.

- New suggested orders.
- Change to existing orders (production or purchase).

Suggested order

The quantity from all colour/size records for the same style and date will be accumulated and a style record created for the total.

Change to existing order

It could be that different colour/sizes on a single order will have had different suggested rescheduling reported, for example:

- Bring in due date.
- Take out due date.
- Change quantity.

The system changes the original order to match the most popular date (that is, the date that most colours/sizes are linked to). It then creates suggested orders for the other dates, changing the original order to suggested for all colours/sizes.

Phase 5: Explode Recommendations

The purpose of this Phase is to generate dependent demand records corresponding to supply changes which have been identified in Phase 3

Suggested & Planned Orders

Each suggested and planned order is firstly backscheduled to determine the planned start date of each operation. Then the materials which are issued at that operation are examined to determine whether they are effective on the specific date. If so, a dependent demand record is created. This is described in the following sections:

- Backscheduling.
- Material effectivity check.

Backscheduling - Establishes material requirement dates.

Using the suggested due date as a basis, the route/BOM is exploded through its operations. The lead time of each operation is calculated using the standard times, and applying the recommended order supply quantity to inflate or deflate the times. The actual route used to do this depends on the Run Parameters screen as follows:

- If Include Capacity is set to 1 (yes), and a route entered in the Capacity Route field, then the entered capacity route will be used.
- If Include Capacity is set to 1 (yes), and Use Planning Route set to 1 (yes) then the item's planning route will be used.
- If Include Capacity is set to 0 (no), then the item's planning route will be used.

Any wastage defined at the operation is also applied to the cycle times to arrive at an operation start date. The lead time calculation takes into account any parallel processing operations, defined by the bundle ticket type. The dates are validated against the calendar defined at each scheduled machine in order to establish a working day for the operation start date.

The routine also creates operation load records (for suggested production orders only); that is, start and end dates, machine used, labour profile used, machine and labour time (run and setting). These records are used in Phase 6.

Backscheduling Example	
Move Time	Specified in route/BOM maintenance.
Queue Time	Specified by machine in machine maintenance.
Duration Time	Calculated from the times specified in the route.
Order Due Date	30/10/95

(A) Order Quantity	10
(B) Standard Lot Size	1
(C) Standard Machine Time	3 hours
(D) Operational Wastage	10%
(E) Machine Duration Time	$((C \times A)/B) \times ((100/(100-D))$ $((3 \times 10)/1) \times ((100/(100-10))$ 33.33 hours
(F) Standard Shift Length	7 hours
Duration Days	$(E)/(F)$ $33.33/7$ 4.76 days 5 days (rounded up)
Start Date	25/10/95

Figure -: Backscheduling Example

Material effectivity check - Having established the planned start date of the operation, the run extracts all materials required at the operation and checks their effectivity.

A requirement is generated if:

- The material's effectivity start date is less than or equal to the calculated operation start date, and *its effectivity stop date is greater than the operation start date*

Effective materials will then be checked for a defined issuing stockroom in the planning model. The quantity of material required is calculated as:

- Recommended order quantity x Cumulative operation wastage x Quantity per standard lot size x Material wastage

Material Requirement Calculation	
A) Order quantity	100
B) Cumulative wastage at	15%

operation	
C) Lot size on planning route	10
D) Quantity per standard lot size	10
E) Material wastage	5%
F) Cumulative operation wastage	$100/(100-B)=100/(100-15)=1.1764$
G) Material wastage	$100/(100-E)=100/(100-5)=1.0526$
Requirement	$A \times F/C \times D \times G$
	$100 \times 1.1764/10 \times 10 \times 1.0526$
	123.8

Figure -: Material Requirement Calculation



If the Fixed Quantity Per flag for a material on a route/BOM is set to 1 (yes), then the requirement is always equal to the defined Quantity Per Lot size; wastage is not calculated

Firm, Released & Active Production Orders

These orders are exploded down to the material requirements defined on production orders.

The current order due date or the run suggested due date if different, is used as the starting point for the back scheduling routine to establish operation start and material requirement dates. The system will also:

- Calculate outstanding operation lead time.
- Calculate material requirements at operations.

Calculate outstanding operation lead time - The outstanding time at an operation is calculated based on the amount of work completed. The operation lead times are examined when back scheduling, rather than applying any fixed lead time.

The operation start dates based on the order quantity are calculated. If the run recommends a change of order quantity, the existing operation lead times are re-calculated based on the revised order quantity.

Calculate material requirements at operations - The re-calculated operation start dates form the revised demand dates for all materials scheduled at an operation. These revised demand dates are used when analysing the demands for the materials. The run translates suggested changes at the parent level into the generated demand at the material requirement level to meet the higher level schedule. This effect is cascaded down the levels of the process structure, ensuring an integrated material plan.

Requirement quantities are recalculated based on suggested order quantity changes at the production order level. They are increased or decreased accordingly.

The requirement quantity outstanding is calculated in the following way:

Quantity required - Quantity issued

If a production order is suggested for cancellation, no explosion is performed, so the material demand for the cancelled order is dropped from the plan. If materials have already been issued, then these should be returned to the issuing stockroom(s)

Purchased Items

Recommendations for purchased items: item types B (bought out), P (purchased) or T (consumable tool); are not exploded through routes, and the review goes no further in dealing with these items.

Lead times for purchased items are calculated as follows:

- For existing orders, the receipt due date is used, except for unscheduled schedule or blanket orders where the due date is taken to be the end date of the run.
- For items for which no purchase orders currently exist, the item/supplier profile in Style Purchase Management is used, if available, if not, the lead time for the item/stockroom is used.

Phase 6: Generate Daily Machine/Labour Loading



This Phase is only included if the Include Capacity flag on the Run Parameters is set to 1 (yes)

A file of operation load for suggested orders generated in Phase 5 is then spread into:

- Daily machine loadings.
- Daily labour loadings.

Daily machine loadings

This will be the operation loadings for machines, but split into individual days.

Daily labour loadings

Using the operation loads, separate records will be generated for each labour skill within the labour profile defined at the operation.

Worked Examples

Worked examples are given below.

MPS Planning

This section comprises examples of an MPS run using different order policy codes. In order to illustrate the calculation being made by an MPS the following example is used: Style X has the following planned parameters:

Style Parameter Table	
Physical Stock	10
Safety Stock	5
Minimum Order Quantity	20
Maximum Order Quantity	50
Multiple Order Quantity	30
Fixed Order Quantity	18
Number Of Days Supply	21
Lead Time	0

Demand								
Period	1	2	3	4	5	6	7	8
Demand	15	40	40	30	15		45	

The following terms are used in this example:

Demand

Net demand depending on the demand policy code

Supply

Total quantity on purchase and/or production order

Available

Quantity available from previous period *plus* supply *less* Demand

Net Req.

Demand *plus* Safety stock *less* Previous period available *less* Existing supplies

Planned Receipt

System suggested order

Planned Available

Stock available if planned orders are taken into account

MPS Planning: Using Order Policy Code A

The planned receipt is always equal to the net requirement.

Period	1	2	3	4	5	6	7	8
--------	---	---	---	---	---	---	---	---

AVAILABLE								
Previous Available	0	-5	-45	-85	-115	-130	-130	-175
Plus Supply	10	0	0	0	0	0	0	0
Total	10	-5	-45	-85	-115	-130	-130	-175
Less Demand	15	40	40	30	15	0	45	0
	-5	-45	-85	-115	-130	-130	-175	-175

NET REQUIREMENT								
Demand	15	40	40	30	15	0	45	0
Plus Safety	5	5	5	5	5	5	5	5
Total	20	45	45	35	20	5	50	5
Less Previous	0	-5	-45	-85	-115	-130	-130	-175
Total	20	50	90	120	135	135	180	180
Less Supply	10	0	0	0	0	0	0	0
	10	50	90	120	135	135	180	180

PLANNED RECEIPT								
Requirement	10	50	90	120	135	135	180	180
Less Made	0	10	50	90	120	135	135	180
Total	10	40	40	30	15	0	45	0
Policy Code Adj.	0	0	0	0	0	0	0	0
	10	40	40	30	15	0	45	0

PLANNED AVAILABLE								
Planned Receipt	10	40	40	30	15	0	45	0
Stock Previous Period	10	5	5	5	5	5	5	5
Total	20	45	45	35	20	5	50	5
Less Demand	15	40	40	30	15	0	45	0
	5	5	5	5	5	5	5	5

MPS Planning: Using Order Policy Code B

The planned receipt is equal to the net requirement or the Minimum Order Quantity, whichever is the smaller.

Period	1	2	3	4	5	6	7	8
--------	---	---	---	---	---	---	---	---

AVAILABLE								
Previous Available	0	-5	-45	-85	-115	-130	-130	-175
Plus Supply	10	0	0	0	0	0	0	0
Total	10	-5	-45	-85	-115	-130	-130	-175
Less Demand	15	40	40	30	15	0	45	0
	-5	-45	-85	-115	-130	-130	-175	-175

NET REQUIREMENT								
Demand	15	40	40	30	15	0	45	0
Plus Safety	5	5	5	5	5	5	5	5
Total	20	45	45	35	20	5	50	5
Less Previous	0	-5	-45	-85	-115	-130	-130	-175
Total	20	50	90	120	135	135	180	180
Less Supply	10	0	0	0	0	0	0	0
	10	50	90	120	135	135	180	180

PLANNED RECEIPT								
Requirement	10	50	90	120	135	135	180	180
Less Made	0	20	50	90	120	140	140	180
Total	10	30	40	30	15	-5	40	0
Policy Code Adj.	10	0	0	0	5	5	0	0
	20	30	40	30	20	0	40	0

PLANNED AVAILABLE								
Planned Receipt	20	30	40	30	20	0	40	0
Stock Previous Period	10	15	5	5	5	10	10	5
Total	30	45	45	35	25	10	50	5
Less Demand	15	40	40	30	15	0	45	0

15	5	5	5	10	10	5	5
----	---	---	---	----	----	---	---

MPS Planning: Using Order Policy Code D

The planned receipt is always a multiple of the Fixed Order Quantity.

Period	1	2	3	4	5	6	7	8
--------	---	---	---	---	---	---	---	---

AVAILABLE								
Previous Available	0	-5	-45	-85	-115	-130	-130	-175
Plus Supply	10	0	0	0	0	0	0	0
Total	10	-5	-45	-85	-115	-130	-130	-175
Less Demand	15	40	40	30	15	0	45	0
	-5	-45	-85	-115	-130	-130	-175	-175

NET REQUIREMENT								
Demand	15	40	40	30	15	0	45	0
Plus Safety	5	5	5	5	5	5	5	5
Total	20	45	45	35	20	5	50	5
Less Previous	0	-5	-45	-85	-115	-130	-130	-175
Total	20	50	90	120	135	135	180	180
Less Supply	10	0	0	0	0	0	0	0
	10	50	90	120	135	135	180	180

PLANNED RECEIPT								
Requirement	10	50	90	120	135	135	180	180
Less Made	0	18	54	90	126	144	144	180
Total	10	32	36	30	9	-9	36	0
Policy Code Adj.	8	4	0	6	9	9	0	0
	18	36	36	36	18	0	36	0

PLANNED AVAILABLE								
Planned Receipt	18	36	36	36	18	0	36	0
Stock Previous Period	10	13	9	5	11	14	14	5
Total	28	49	45	41	29	14	50	5
Less Demand	15	40	40	30	15	0	45	0
	13	9	5	11	14	14	5	5

MPS Planning: Using Order Policy Code G

The planned receipt is sufficient to satisfy the forward demand for the *Number of Days Supply* from the requirement date.

Period	1	2	3	4	5	6	7	8
--------	---	---	---	---	---	---	---	---

AVAILABLE								
Previous Available	0	-5	-45	-85	-115	-130	-130	-175
Plus Supply	10	0	0	0	0	0	0	0
Total	10	-5	-45	-85	-115	-130	-130	-175
Less Demand	15	40	40	30	15	0	45	0
	-5	-45	-85	-115	-130	-130	-175	-175

NET REQUIREMENT								
Demand	15	40	40	30	15	0	45	0
Plus Safety	5	5	5	5	5	5	5	5
Total	20	45	45	35	20	5	50	5
Less Previous	0	-5	-45	-85	-115	-130	-130	-175
Total	20	50	90	120	135	135	180	180
Less Supply	10	0	0	0	0	0	0	0
	10	50	90	120	135	135	180	180

PLANNED RECEIPT								
Requirement	10	50	90	120	135	135	180	180
Less Made	0	90	90	90	135	135	135	180
Total	10	-40	0	30	0	0	45	0
Policy Code Adj.	80	40	0	15	0	0	0	0
	90	0	0	45	0	0	45	0

PLANNED AVAILABLE								
Planned Receipt	90	0	0	45	0	0	45	0
Stock Previous Period	10	85	45	5	20	5	5	5
Total	100	85	45	50	20	5	50	5
Less Demand	15	40	40	30	15	0	45	0

85	45	5	20	5	5	5	5
----	----	---	----	---	---	---	---

MPS Planning: Using Order Policy Code H

The planned receipt is equal to the *Minimum Order Quantity* plus a number of *Multiple Order Quantities* sufficient to satisfy the net requirement.

Period	1	2	3	4	5	6	7	8
--------	---	---	---	---	---	---	---	---

AVAILABLE								
Previous Available	0	-5	-45	-85	-115	-130	-130	-275
Plus Supply	10	0	0	0	0	0	0	0
Total	10	-5	-45	-85	-115	-130	-130	-275
Less Demand	15	40	40	30	15	0	145	0
	-5	-45	-85	-115	-130	-130	-275	-275

NET REQUIREMENT								
Demand	15	40	40	30	15	0	145	0
Plus Safety	5	5	5	5	5	5	5	5
Total	20	45	45	35	20	58	150	5
Less Previous	0	-5	-45	-85	-115	-130	-130	-175
Total	20	50	90	120	135	135	280	280
Less Supply	10	0	0	0	0	0	0	0
	10	50	90	120	135	135	280	280

PLANNED RECEIPT								
Requirement	10	50	90	120	135	135	280	280
Less Made	0	30	60	90	120	150	150	300
Total	10	20	30	30	15	-15	130	-20
Policy Code Adj.	20	10	0	0	15	15	20	20
	30	30	30	30	30	0	150	0

PLANNED AVAILABLE								
Planned Receipt	30	30	30	30	30	0	150	0
Stock Previous Period	10	25	15	5	5	20	20	25
Total	40	55	45	35	35	20	170	25
Less Demand	15	40	40	30	15	0	145	0

25	15	5	5	20	20	25	25
----	----	---	---	----	----	----	----

In period 7 the demand has been increased to 145 to show the effect of exceeding the maximum quantity.

Demand, Supply And Action Codes

MPS and MRP reviews and reports display/print status codes which identify recommendations:

- Demand status codes.
- Supply status codes.
- Action codes.

These codes have been summarised in the following tables. They are defined on the Parameter file, accessed from the Style Production, Utilities menu.



Note: Codes which include the term (Excluded) relate to demand or supply which is excluded from the review.

For example, if a sales forecast (30 items) and sales order demand (60 items) exist on a particular day, and the demand policy utilises the greater of actual demand or forecast, then the 60 will be shown with a status of SO and an FX against the 30 forecast as it is not used (excluded).

Codes which include the term STYLE relate to demand which is at style level only; that is, there is no defined requirement to style/colour/size.

Demand Status Codes

Demand status codes are defined on the Parameter file against type PDSC.

Demand Status Codes	
CA	Cancelled
CD	Cumulative demand, if cumulative consumption of forecast
CW	Confirmed production order
FC	Sales forecast
FS	Stock forecast
FX	Forecast (excluded)
IW	Active production order
MA	[MPS] Manual adjustment
MX	[MPS] Manual adjustment (excluded)
PW	Planned production order
PX	Suggested purchase order (excluded)
RW	Released production order
SO	Sales order
SP	Suggested purchase order
SW	Suggested production order
SX	Sales order (excluded)
WX	Production order (excluded)
YF	STYLE sales forecast
YP	STYLE suggested purchase order
YS	STYLE stock forecast
YW	STYLE suggested production order

Figure -: Demand Status Codes

Supply Status Codes

Supply status codes are defined on the parameter file, against type PSSC

Supply Status Codes	
CN	Cancelled
CO	Complete production order
CP	Confirmed purchase
CW	Confirmed production order
GI	Goods inwards
IS	In inspection
IW	Active production order
PW	Planned production order
RW	Released production order
SP	Suggested purchase order
SW	Suggested production order
YW	STYLE suggested production order
YP	STYLE suggested purchase order

Figure -: Supply Status Codes

Action Codes

The major purpose of the MPS/MRP runs is to provide planning support. The advice given is based on the latest status of material requirements and the current order status.

The following table indicates the recommendations that may be made. These codes may appear in combination; and examples of possible suggestions are given subsequently to illustrate the sort of actions which may be suggested.

Action codes are defined on the Parameter file, against type PEXC.

Material Effectivity

If as a result of a recommendation, material effectivity is compromised, that is, the material is not effective on the route/BOM on that date, then warning codes are also displayed (these are included in the table).

Reschedule 'In' Or 'Out'

Expedite or defer. References to reschedule 'in' indicates that the run is suggesting that the due date is brought forward ('into' the schedule) or expedited.

References to reschedule 'out' indicates that the run is suggesting that the due date is deferred ('out' from the schedule).

The *Review* facilities for both MPS and MRP enable you to view orders in both original and suggested due date sequence.



Codes shown in the table as comprising 4 consecutive characters, for example ROCQ, will be displayed as individual elements separated by a date, and with a suggested change quantity where applicable. For example RO11/09 CQ 2 .

The examples given following the table will help clarify this.

Action Codes	
CA	Cancel order
CQ	Change quantity (mix)
EF	Check effectivity
EI	Reschedule 'in' and check effectivity
EICQ	Reschedule 'in' and change quantity and check effectivity
EO	Reschedule 'out' and check effectivity
EOCQ	Reschedule 'out' and change quantity and check effectivity
EQCQ	Change quantity and check effectivity
RI	Reschedule 'in'
RICQ	Reschedule 'in' and change quantity
RO	Reschedule 'out'

ROCQ	Reschedule 'out' and change quantity
------	--------------------------------------

Figure -: Action Codes

Action Recommendation Examples			
Order	Quantity	St	Action
W000156	122	CW	RI 15/08 CQ 1180
Reschedule order due date 'in' to 15 August, and change quantity from 122 to 1180			
W0007716	30	IW	RO11/09 CQ 25
Reschedule order due date 'out'; defer until 11 September, and change quantity from 30 to 25			

Figure -: Action Recommendation Examples

MPS Specific Processes

- Enables capacity planning.
- Selects MPS items (planning type 1).
- Enables amendments to be made to suggested production orders; allowing changes in quantities, routes, due dates and planning filters prior to confirmation.
- Allows a cut-off low level code to be defined as a parameter; below which the run will not look for any occurrence of MPS items.
- Enables different versions of the master production schedule to be saved against identifying revision level codes.
- Enables demand to be reviewed part way through the run; to allow you to effect changes which will be incorporated when the run commences.

Capacity Planning

This is a unique function within Master Production Scheduling and enables you to compare available machine and labour capacity against projected work loads.

This can either be carried out using the standard planning route/BOMs or specially derived capacity routes.

A Capacity Review function then enables you to change the work load schedule at production order level.

MRP Specific Processes

- Selects MRP items (planning type 0).
- Enables a selective MRP run based on a range of items rather than all items in the planning model.
- Enables MPS suggestions to be used as direct input to the material plan.